THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The cell was first observed and named by: 1

(a) Robert Hooke **(b)** Rudolf Virchow
(c) Robert Brown **(d)** Anton van Leeuwenhoek

2. A group of similar cells performing a specific function is called a: 1

(a) tissue **(b)** organelle **(c)** organism **(d)** organ

3. The variety of living organisms found on the Earth is called: 1

(a) ecology **(b)** taxonomy **(c)** heredity **(d)** biodiversity

4. Reproduction in which a single parent produces offspring identical to itself is: 1

(a) asexual reproduction **(b)** sexual reproduction
(c) fertilisation **(d)** pollination

5. The four spheres of the Earth system are the geosphere, hydrosphere, atmosphere and: 1

(a) lithosphere only **(b)** biosphere **(c)** stratosphere **(d)** troposphere

6. A cell without a well-defined nucleus is a: 1

(a) eukaryotic cell **(b)** prokaryotic cell **(c)** animal cell **(d)** plant cell

7. The tissue that makes the husk of a coconut is the: 1

(a) parenchyma **(b)** sclerenchyma **(c)** phloem **(d)** collenchyma

8. The simplest plants with a thallus-like body are: 1

(a) ferns **(b)** mosses **(c)** gymnosperms **(d)** algae

9. The male reproductive part of a flower is the: 1

(a) stamen **(b)** sepal **(c)** petal **(d)** carpel

10. The surface that reflects the most solar radiation is: 1

(a) dark soil **(b)** ocean **(c)** fresh snow **(d)** forest

11. The endoplasmic reticulum with ribosomes attached is the: 1

(a) vacuole **(b)** Golgi **(c)** smooth ER **(d)** rough ER

12. Animal tissue that covers or lines the body surfaces is: 1

(a) connective tissue **(b)** nervous tissue
(c) muscular tissue **(d)** epithelial tissue

13. Sponges belong to the phylum: 1

(a) Mollusca **(b)** Annelida **(c)** Porifera **(d)** Cnidaria

14. After fertilisation, the ovary develops into a: 1

(a) anther **(b)** fruit **(c)** seed **(d)** flower

15. The ozone layer is present in the: 1

(a) stratosphere **(b)** troposphere **(c)** geosphere **(d)** mesosphere

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Who discovered the cell? What did he observe? 2

17. What is a tissue? Why do multicellular organisms need tissues? 2

18. What is biodiversity? Why is classification necessary? 2

19. Define reproduction. Why is it important? 2

20. Name the four spheres of the Earth system and state one feature of each. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A raisin is placed in water and another in a strong sugar solution. Predict and explain what happens. 3

22. Explain with an activity how roots grow at their tips. 3

23. Name any four characteristics used for classifying organisms. 3

24. Describe the binary fission in Amoeba with the steps. 3

OR

Explain how the four spheres interact with an example of a rainy day. 3

25. Explain why red blood cells and nerve cells have different shapes. 3

OR

Describe the structure of a neuron with a labelled description. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the structure of a typical animal cell with a labelled description of the organelles and their functions. 5

OR

Describe the meristematic tissues: types, location and functions with the way they differ from permanent tissues. 5

27. Explain the five-kingdom classification with the basis, characters and examples of each kingdom. 5

OR

Describe the different methods of asexual reproduction in animals and plants with examples. 5

END OF QUESTION PAPER — Total: 50 Marks

UNDER PRINCIPAL